

RioClear Initiative: Integrated Surface Water Contamination and Drinking Water Compliance Surveillance Along the U.S.–Mexico Border

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Introduction

Across the 13 Texas border counties along the Rio Grande there is over 2.87 million residents that rely on public water systems and adjacent surface-water resources. Our project objective is to synthesize regulatory data from the U.S. EPA and water sample data from the USGS/TCEQ into a unified, open-source dashboard to locate failing drinking-water systems, identify surface-water contamination hotspots, and compute an objective priority score to guide intervention.

What are we looking for?

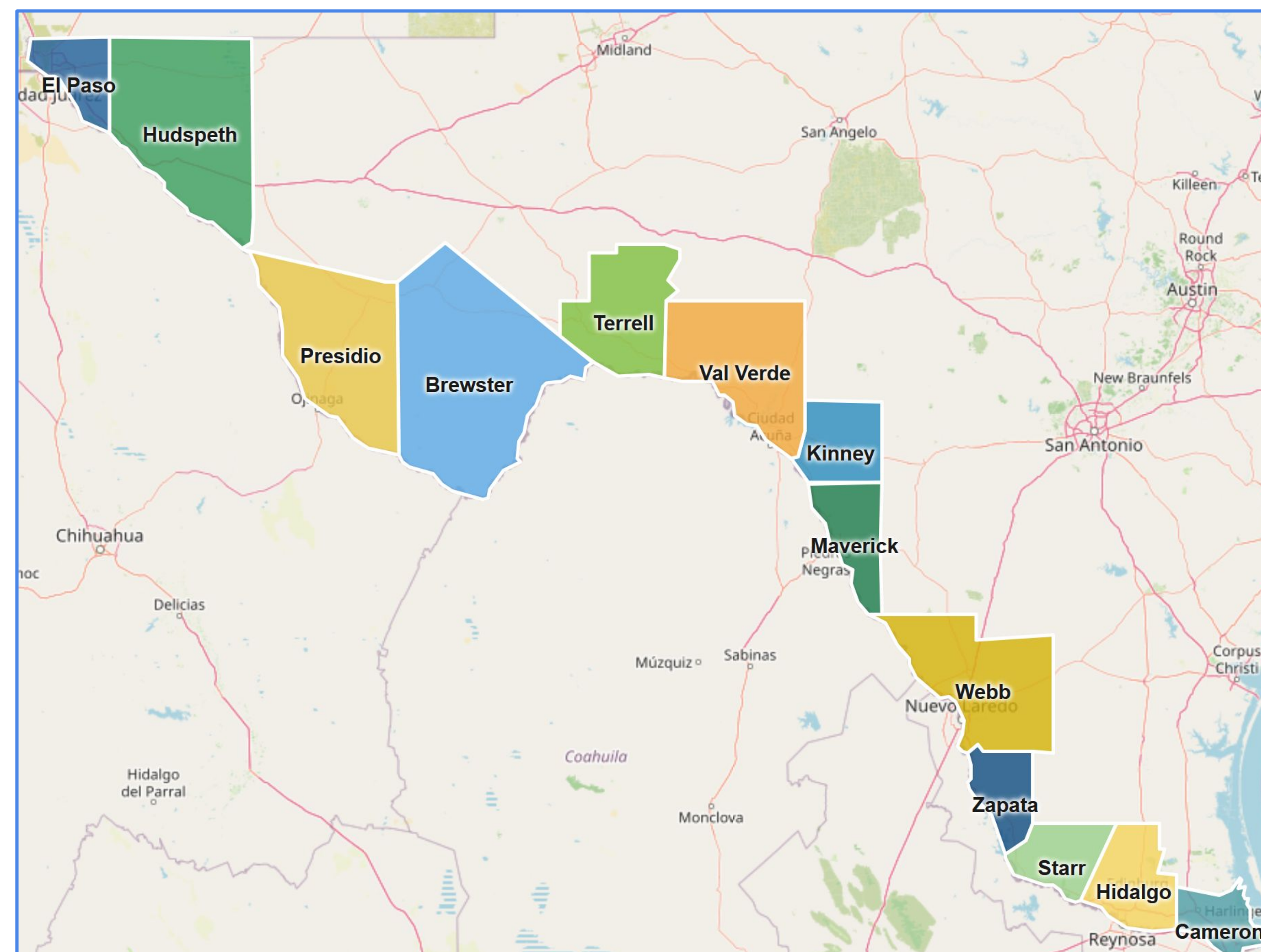
- Where are municipal drinking-water systems experiencing health-based compliance failures along the Rio Grande border region?
- What is the spatial distribution and severity of microbial (*E. coli*) and chemical contamination in surface waters?
- Which border counties present the highest cumulative environmental health risks requiring immediate action?

Methodology and Data Sources

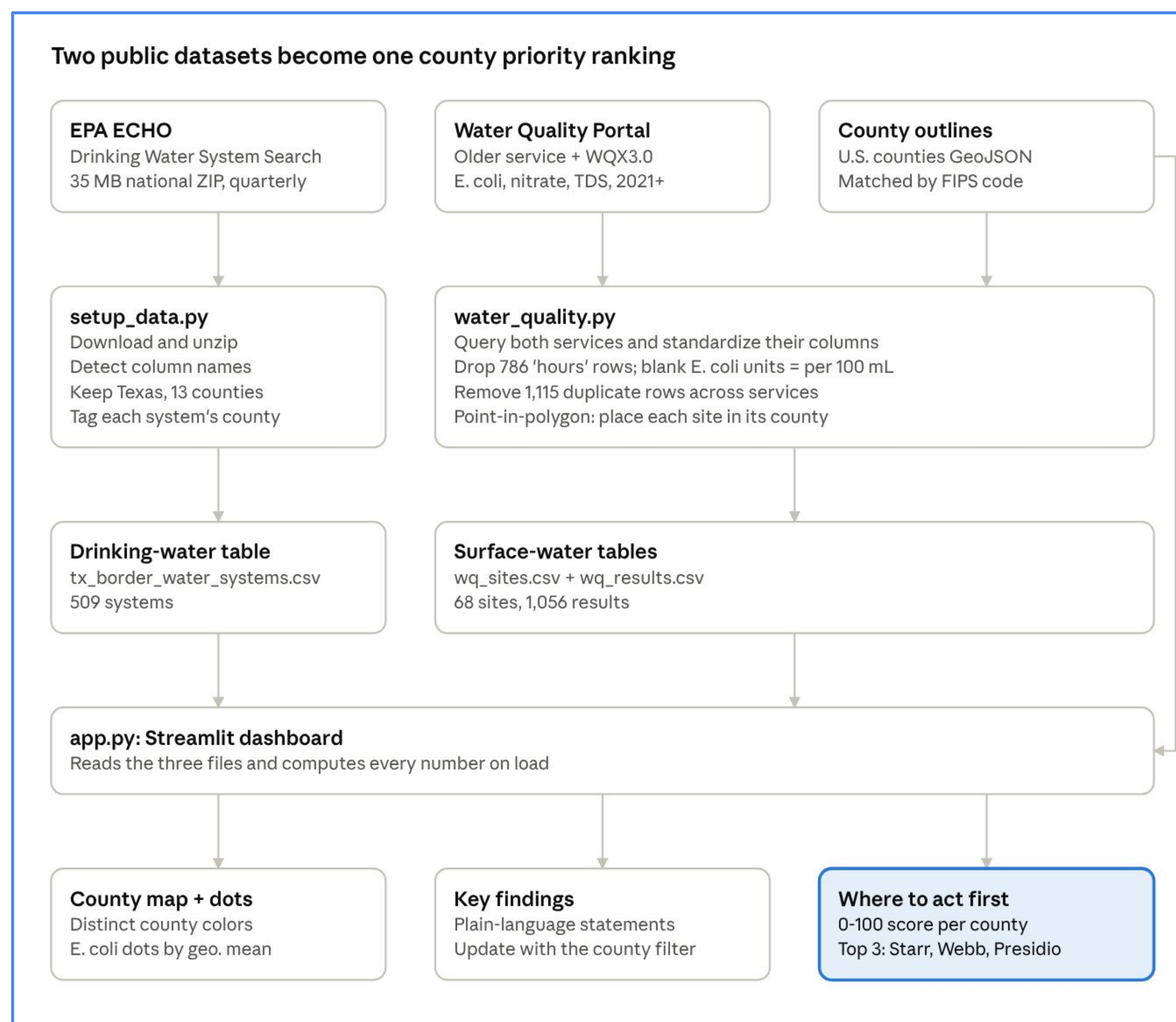
- EPA ECHO:** compliance records for 509 drinking-water systems.
- E. coli thresholds:** 126 per 100 mL (geometric mean) is the EPA/Texas level for safe swimming and wading; 399 is the Texas single-sample level.
- Water Quality Portal (USGS + EPA):** *E. coli*, nitrate, and TDS samples from 68 sites, 2021–2026.
- Merging two services:** combined the portal's older service and its newer WQX3.0 service, which holds USGS data since March 2024, and removed 1,115 duplicate rows.
- Placing sites:** a point-in-polygon test assigns each site to its county from its coordinates.
- Priority score (0–100):** the average of three signals, each scaled from lowest county to highest: share of health-flagged systems, serious violations per 10 systems, and share of *E. coli* sites above 126.
- Tools:** Python, pandas, Streamlit, Folium. Each site is placed in its county by its coordinates.

Rank / County	Priority Score	Primary Risk Driver	Health-Flagged Systems	Serious Violators	Population Affected	<i>E. coli</i> Sites > 126	Worst Site (<i>E. coli</i> Geom. Mean)
1. Starr	76	Health-Flagged Systems	4 / 22	2	18,918	2 / 2	2,151
2. Webb	39	Surface <i>E. coli</i> Exceedance	2 / 40	4	871	3 / 5	8,172
3. Presidio	38	Health-Flagged Systems	3 / 28	2	252	1 / 3	139

Rio Grande Counties Map



Data Pipeline and Integration



Acknowledgments

Thank you! Rowdy Hacks XII for providing the platform and opportunity to challenge ourselves, expand our technical capabilities, and apply our skills in a collaborative environment.

Key Findings

Across the 13 counties, 23 drinking-water systems serving about 86,231 people carry health-based violation flags, and 24 of 57 surface-water sites exceed the *E. coli* level for safe swimming and wading.

Drinking water (EPA ECHO)

- 509 public water systems serve 2,872,711 people.
- 23 systems carry a health-based violation flag; they serve about 86,231 people.
- 14 systems are listed as serious violators.
- Starr has the highest share of flagged systems: 18.2% (4 of 22).

Rivers and creeks (water samples, 2021–2026)

- E. coli* comes from sewage and animal waste. Above 126 is unsafe for swimming or wading.
- 24 of 57 sites tested (about 4 in 10) were above the safe level.
- Rio Grande below Laredo: about 65 times the safe level, the worst site we found.
- Los Olmos Creek (Starr County): about 17 times the safe level.
- Webb County (Laredo) has the most contaminated water overall.
- Hudspeth and Kinney counties have had no *E. coli* testing since 2021.

What We Learned

- Real-world data is messy.** Our first run returned zero *E. coli* results. Printing diagnostics showed 1,218 rows with blank units and 786 lab records in "hours." Fixing the unit rules recovered 450 results.
- Geospatial analysis:** working with FIPS codes, GeoJSON county outlines, and point-in-polygon tests to place 68 sampling sites in their counties.
- Public data services:** querying the Water Quality Portal's older and newer (WQX3.0) services and merging them. The older service stopped receiving new USGS data in March 2024.
- Honest interpretation:** high river bacteria levels are not the same as unsafe tap water, so we present the results as a screening tool, not an official assessment.
- Building and deploying** a live dashboard with Streamlit and Folium, from local prototype to public web app in one weekend.

Recommendations And Action Items

- Targeted Compliance Support:** Prioritize regulatory intervention and infrastructure funding for health-flagged drinking water systems in Starr and Presidio counties.14
- Source Tracking Investigations:** Initiate microbial source tracking along the Rio Grande near Laredo and Los Olmos Creek to differentiate agricultural run-off from municipal wastewater overflows.14
- Address Monitoring Deficits:** Re-establish active surface-water sampling stations in Hudspeth and Kinney counties, which currently lack \$*E. coli*\$ monitoring data since 2021.